

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY (A)

RVS NAGAR, CHITTOOR

II B.Tech-II Semester:

Subject name: – Optimization Technique

Branch: AIML, DS, IT, CAI

Sessional Examination-I

Subject code: 23AME05

Max Marks: 25

Time: 2 Hours

Part A

Answer all the questions, each question carry equal marks

5 x 2 = 10 M

Q. NO	QUESTION	CO	BL
1	What is a feasible solution?	CO1	L1
2	State any two Assumptions in Linear Programming.	CO1	L1
3	Write the condition for converting an LPP into standard form with Example?	CO1	L2
4	Give the different practical application of transportation problem?	CO2	L1
5	What is the Hungarian algorithm?	CO2	L1

Part B

Answer all the questions, each question carry equal marks

3 x 5 = 15 M

Q. NO	QUESTION	CO	BL
6.(a)	Graphical Method – Minimization Solve the following LPP using the graphical method: Maximize $Z=6X_1+8X_2$ Subject to: $5X_1+10X_2 \leq 60$ $4X_1+4X_2 \leq 40$ X_1 and $X_2 \geq 0$	CO1	L2
	OR		
6.(b)	Solve the following LP model and solve it by using the Big M method: Minimize $Z=10X_1+15X_2+20X_3$ Subject to: $2X_1+4X_2+6X_3 \geq 24$ $3X_1+9X_2+6X_3 \geq 30$ $X_1, X_2, X_3 \geq 0$	CO1	L2
7.(a)	Solve the following LPP using the Two-Phase method: Maximize $Z=3x_1+2x_2$ Subject to: $x_1+x_2=4$ $2x_1+x_2 \geq 5$ $x_1, x_2 \geq 0$	CO1	L2
	OR		
7.(b)	Consider the following transportation problem involving three sources and four destinations. The cell entries represent the cost of transportation per unit. Obtain the initial, basic feasible solution using the following methods (a) Northwest corner cell method (b) least cost cell method (c) VAM method	CO2	L2

	D1	D2	D3	Supply
S1	20	10	15	200
S2	10	12	9	300
S3	25	30	18	500
Demand	200	400	400	1000

- 8.(a) Solve the following assignment problem using Hungarian method. The matrix entries are processing times in hours

Operator

Job	1	2	3	4	5
1	20	22	35	22	18
2	4	26	24	24	7
3	23	14	17	19	19
4	17	15	16	18	15
5	16	19	21	19	25

71 hours

OR

- 8.(b) Consider the assignment problem as shown below. In this problem, five different jobs are to be assigned to five different operators such that the total processing time is minimized. The matrix entries represent processing times in hours. CO2 L2

		Operator				
		1	2	3	4	5
Job	1	5	6	8	6	4
	2	4	8	7	7	5
	3	7	7	4	5	4
	4	6	5	6	7	5
	5	4	7	8	6	8

II B.Tech-II Semester:

Subject name: – Probability & Statistics

Branch: CSE, CSM, IT, CAI, IOT & CSBS

Sessional Examination-I

Subject code: 23AHS21

Time: 2 Hours

Max Marks: 25

Part A

Answer all the questions, each question carry equal marks

5 x 2 = 10 M

Q. NO	QUESTION	CO	BL
1	Define rank correlation.	CO1	L1
2	State one difference between mean and mode.	CO1	L1
3	Calculate the arithmetic Mean of the following data. 40,50,55,78,58,60.	CO1	L2
4	Define discrete and continuous random variables.	CO2	L2
5	A card is drawn from a well shuffled pack of cards. What is the probability that it is either a spade or an ace.	CO2	L2

Part B

Answer all the questions, each question carry equal marks

3 x 5 = 15 M

Q. NO	QUESTION	CO	BL
6.(a)	Calculate the median from the following data	CO1	L3

u3.47

Marks	10-25	25-40	40-55	55-70	70-85	85-100
Frequency	6	20	44	26	3	1

OR

6.(b) Obtain the rank correlation coefficient for the following data

X	68	64	75	50	64	80	75	40	55	64
Y	62	58	68	45	81	60	68	48	50	70

7.(a) Find the variance and Standard deviation for the given Data.

u3.58

x	6	10	14	18	24	28	30
f	2	4	7	12	8	4	3

OR

7.(b) The continuous probability function $f(x) = kx^2 e^{-x}$ when $x \geq 0$. Find (i) k (ii) Mean (iii) Variance.

8.(a) A bag contains 8 red and 6 blue balls. Two drawing of each 2 balls are made. Find the probability that the first drawing gives two red balls and second drawing gives 2 blue balls, if the balls drawn first are replaced before the second draw.

OR

8.(b) Find the constant k such that $f(x) = kx^2, 0 < x < 3$ is a probability function. Find the distribution function $F(x), P(1 < X \leq 2)$.

Table

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY (A)
RVS NAGAR, CHITTOOR

II B.Tech-II Semester:

Subject name: – DATABASE MANAGEMENT SYSTEMS

Branch: CSM, DS, IT, CAI & CSBS

Time: 2 Hours

Sessional Examination-I

Subject code: 23ACS12a

Max Marks: 25

Part A

Answer all the questions, each question carry equal marks

5 x 2 = 10 M

Q. NO	QUESTION	CO	BL
1	Define a Database system.	CO1	L2
2	Define schema and instance in DBMS.	CO1	L2
3	What is an entity set in the ER model?	CO1	L1
4	Define an attribute with an example.	CO2	L1
5	What is referential integrity constraint?	CO2	L2

Part B

Answer all the questions, each question carry equal marks

3 x 5 = 15 M

Q. NO	QUESTION	CO	BL
6.(a)	Describe various types of database users and explain their roles with examples.	CO1	L4
	OR		
6.(b)	Explain the three-tier schema architecture with a neat diagram and show how it supports data independence.	CO1	L4
7.(a)	Explain the Entity Relationship (ER) model and its components such as entities, attributes, relationships, and constraints.	CO1	L4
	OR		
7.(b)	Explain NULL values. Discuss their importance and problems associated with them	CO2	L2
8.(a)	Explain various relational algebra operations with examples.	CO2	L4
	OR		
8.(b)	Explain domain relational calculus. How is it different from tuple relational	CO2	L3

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY (A)
RVS NAGAR, CHITTOOR

II B.Tech-II Semester:

Subject name: - Machine Learning

Branch: CSM & CAI

Sessional Examination-I

Subject code: 23ACA02

Time: 2 Hours

Max Marks: 25

Part A

Answer all the questions, each question carry equal marks

5 x 2 = 10 M

Q. NO	QUESTION	CO	BL
1	What is Machine Learning?	CO1	L1
2	Mention any two types of data used in Machine Learning.	CO1	L1
3	What is the role of search in Machine Learning?	CO1	L1
4	Define proximity measure.	CO2	L1
5	What is classifier performance?	CO2	L1

Part B

Answer all the questions, each question carry equal marks

3 x 5 = 15 M

Q. NO	QUESTION	CO	BL
6.(a)	Describe the different paradigms of Machine Learning in detail.	CO1	L1
OR			
6.(b)	Explain the process of data acquisition and data representation.	CO1	L1
7.(a)	Explain the complete Machine Learning workflow from data collection to prediction.	CO1	L1
OR			
7.(b)	Explain Euclidean, Manhattan, and Minkowski distance measures.	CO2	L2
8.(a)	Explain the K-Nearest Neighbor (KNN) classifier with steps and example.	CO2	L1
OR			
8.(b)	Explain the performance measures for regression algorithms.	CO2	L3

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY (A)
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II B.Tech-II Semester:

Subject name: -Digital Logic & Computer Organization

Branch: CSM, DS & CAI

Sessional Examination-I

Subject code: 23AEC06

Time: 2 Hours

Max Marks: 25

Part A

Answer all the questions, each question carry equal marks

5 x 2 = 10 M

Q. NO	QUESTION	CO	BL
1	Define the binary number system and its base.	CO1	L1
2	What are the main components of an IEEE-754 single-precision floating-point number?	CO1	L1
3	What is a decoder?	CO1	L1
4	What is a shift register, and give one use case?	CO1	L1
5	What distinguishes system software from application software?	CO1	L2

Part B

Answer all the questions, each question carry equal marks

3 x 5 = 15 M

Q. NO	QUESTION	CO	BL
6.(a)	Convert the following to decimal and then to binary. a) 4234_{16} b) $125F_{16}$ c) 10010011_2 d) 10111111_2	CO1	L3
OR			
6.(b)	Reduce the following function using K-map Technique. i) $f(a,b,c,d) = \sum m(0,1,4,8,9,10)$ ii) $F(A,B,C,D) = \prod M(0,2,4,5,6,7,8,10,13,15)$	CO1	L3
7.(a)	Design a 3-to-8 decoder, and verify its truth table.	CO1	L3
OR			
7.(b)	Discuss Briefly about the operational concepts, Functional units and performance of basic computers.	CO1	L2
8.(a)	Explain about the synchronous and asynchronous counters in detail.	CO1	L2
OR			
8.(b)	Explain how bus structures work in a CPU system, detailing data, address, and control buses.	CO1	L2

8421

A-10
B-11
C-12
D-13
E-14
F-15